



CAVe-Lite

Instruction Set



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List of Abbreviations

API	application programming interface
ASN.1	Abstract Syntax Notation One
BSM	basic safety message
C-V2X	cellular vehicle-to-everything
CAV	connected and automated vehicle
CAVe	Connected and Automated Vehicle education
CV	connected vehicle
DHCP	Dynamic Host Configuration Protocol
DMS	dynamic message sign
DSRC	dedicated short-range communications
FHWA	Federal Highway Administration
GNSS	Global Navigation Satellite System
GPS	Global Positioning System
GUI	graphical user interface
HDMI	High-Definition Multimedia Interface
HIL	hardware-in-the-loop
IP	Internet Protocol
ITIS	International Traveler Information System
ITS	intelligent transportation system
ITS JPO	Intelligent Transportation Systems-Joint Program Office
JSON	Java Script Object Notation



LTE	Long-Term Evolution
NTCIP	National Transportation Communications for ITS Protocol
OBU	onboard unit
OS	operating system
PoE	Power over Ethernet
PSID	Provider Service Identifier
PSM	personal safety message
RSU	roadside unit
SAE	Society of Automotive Engineers
SNMP	Simple Network Management Protocol
SPaT	Signal Phase and Timing
SRM	Signal Request Message
SSH	Secure Shell
STOL	Saxton Transportation Operations Laboratory
TCP/IPv4	Internet Protocol version 4
TFHRC	Turner-Fairbank Highway Research Center
TIM	traveler information message
TMC	traffic management center
TSC	traffic signal controller
UDP	User Datagram Protocol
UPER	unaligned packet encoding rules
USDOT	United States Department of Transportation



V2I	vehicle-to-infrastructure
V2X	vehicle-to-everything
WAVE	Wireless Access in Vehicular Environments
WSA	WAVE Service Advertisement



Wireless Connection

The IPv4 address of any computer can be changed using a graphical user interface (GUI) or the command line. Methods for changing the IP address of a computer vary for every kind of operating system (OS) and even between Linux distributions. IPv4 addresses for computers using CAVe-Lite must be set to static and not dynamic/DHCP. Instructions on changing a computer's IP address can be found from the specific distributor's website. The following steps are for the GUI using Ubuntu 20.04 (64-bit).

Note: Please be aware that attempting to connect to a wireless network managed by your organization may not work. Attempt on a home network, hotspot, or set up your own wireless network using a standard Wi-Fi Router.

1. Follow the instruction at the official Ubuntu Documentation site to connect to your wireless network: <https://help.ubuntu.com/20.04/ubuntu-help/net-wireless-connect.html.en>
2. After connecting to your network, follow these steps to manually set your network settings: <https://help.ubuntu.com/20.04/ubuntu-help/net-manual.html.en>
 - a. Take a screenshot of your assigned IP configurations before manually setting IPv4 configurations. Refer to figure 1.

Details	Identity	IPv4	IPv6	Security
Link speed	1000 Mb/s			
IPv4 Address	192.168.86.44			
IPv6 Address	fe80::9419:f8e7:a9e3:4c2			
Hardware Address	08:00:27:FE:AE:93			
Default Route	192.168.86.1			
DNS	192.168.86.1			

Source: FHWA.

Figure 1. Screenshot of assigned IP configurations

3. Continue with manual IP configurations, ensuring to replace the * below with the number previously assigned.
 - a. Infrastructure Kit: Set the IPv4 IP Address to 192.168.*.146
 - i. Example: 192.168.86.146
 - b. Mobile Kit: Set the IPv4 IP Address to 192.168.*.110
 - i. Example: 192.168.86.110
 - c. Set the Netmask for both kits to 24
 - d. Set the Gateway for both kits to 192.168.*.1
 - i. Example: 192.168.86.1
 - e. Refer to figure 2.



Details Identity **IPv4** IPv6 Security

IPv4 Method

☐ Automatic (DHCP) ☐ Link-Local Only

☒ Manual ☐ Disable

☐ Shared to other computers

Addresses

Address	Netmask	Gateway	
192.168.86.146	24	192.168.86.1	

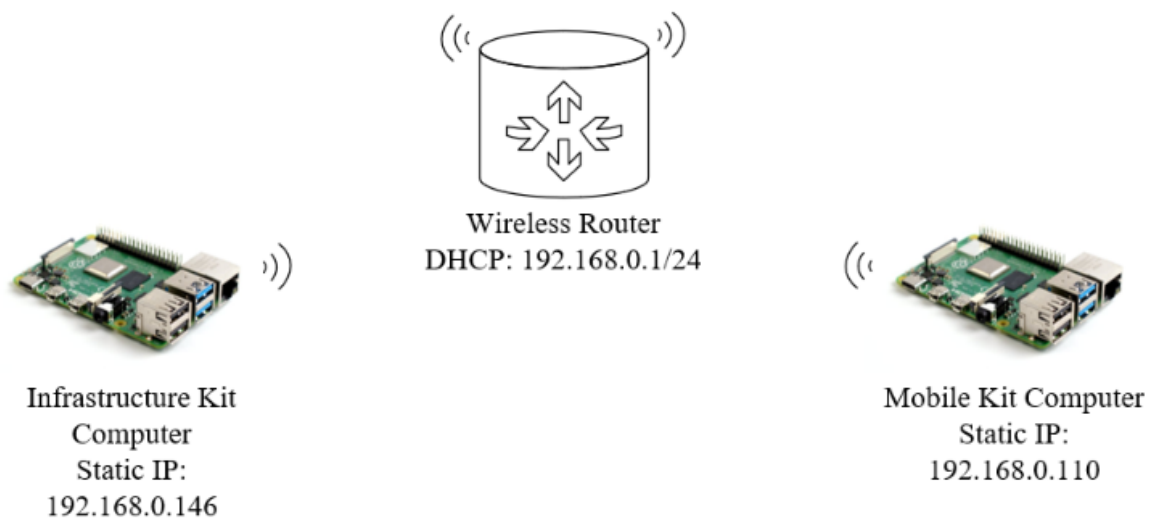
DNS Automatic ☒

Separate IP addresses with commas

Source: FHWA.

Figure 2. Example Infrastructure Kit Static IPv4 Configuration.

4. Apply the configurations, then turn off/on Wi-Fi.
5. Once both kits are connected to the same wireless network, open a terminal and attempt to ping each.
 - a. From infrastructure kit: `ping 192.168.*.110`
 - b. From Mobile Kit: `ping 192.168.*.146`
 - c. There will be a response from each kit. Use <Ctrl-C> to stop the ping.
6. Wireless connection is now complete.



Source: FHWA.

Figure 3. Diagram. CAVe-Lite network configuration.



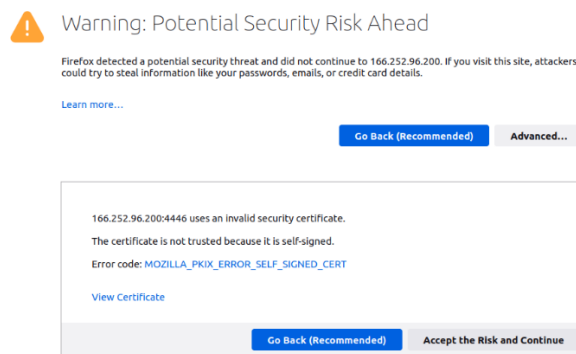
V2X Hub Setup

V2X Hub is available as a Docker image that may be run in a containerized environment (FHWA, 2024). The primary requirement for running V2X Hub is a local computer that can support Docker. Having root access to the device is preferred. For the development of this documentation, a Ubuntu 20.04 was used.

V2X Hub is only installed on the same computer running the **infrastructure kit** scripts. It is recommended to configure the computer to use **192.168.*.146** as the static IPv4 address to be able to mimic the configurations stated within this document.

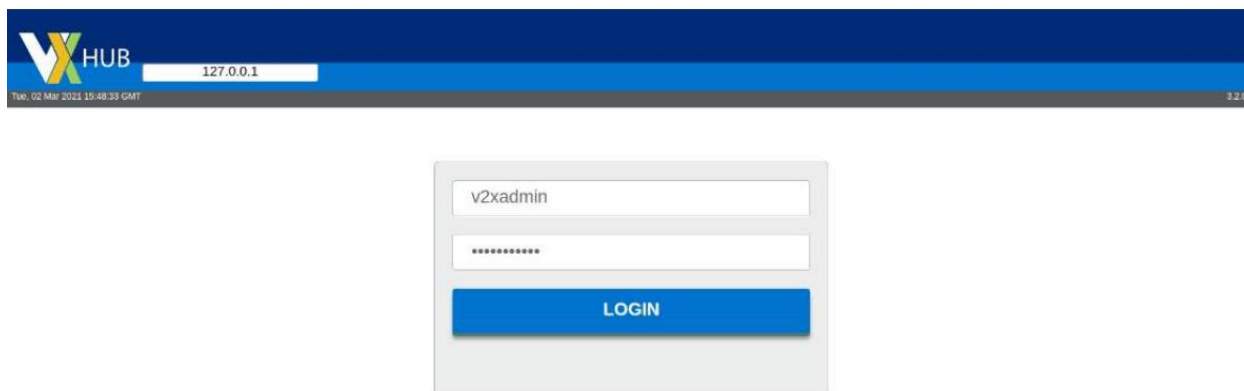
Further documentation for AMD or ARM 64-bit architecture processors are available through the V2X Hub [GitHub repository](#) and [Confluence page](#). Installation can be completed using the following steps, although they may be subject to change:

1. Open a terminal to start download and installation.
2. Download the repository to your Linux computer's home directory:
 - a. `cd`
 - b. `git clone https://github.com/usdot-fhwa-OPS/V2X-Hub.git`
3. Once downloaded, navigate to a directory corresponding to your computer's processor (a) or (b):
 - a. `cd ~/V2X-Hub/configuration/arm64/`
 - b. `cd ~/V2X-Hub/configuration/amd64/`
4. Currently in `~/V2X-Hub/configuration/amd64/`, run the initialization script:
 - a. `sudo ./initialization.sh`
5. Follow the prompts during installation.
 - a. You will be prompted to create a `mysql_password` and `mysql_root_password`. You may make the passwords whatever you'd like, but you will need to remember them.
 - i. **Example:** `ivp`
 - b. You will also be prompted to create a V2X Hub username and password. You may make these whatever you'd like but will need to use them to log into the web UI. Example:
 - i. **Username:** `v2xadmin`
 - ii. **Password:** `V2xHub#321`
 - c. You will then need to enter the `mysql_password` you created in step 5a:
 - i. **Example:** `ivp`
6. After installation is complete, the script will automatically open a web browser with two tabs.
 - a. Navigate to the tab labeled as "Privacy Error" and select the "Advanced" button.
 - b. Click on "proceed to 127.0.0.1 (unsafe)". Refer to figure 4.
 - i. **Note:** This page will not do anything when clicking proceed.
7. Close the Privacy Error tab and wait for the initial V2X Hub tab to finish loading.
8. Enter the login credentials you created in step 5b and login. Refer to figure 5.



Source: FHWA.

Figure 4. Screenshot. Accept Security Credentials.



Source: FHWA.

Figure 5. Diagram. V2X Hub Login Page.

This document will address basic configuration for the V2X Hub plugins. For further reading, please visit the V2X Hub GitHub repository and Confluence page.



CAVe-Lite Setup and Test

Hardware Requirements

- CAVe-Lite Infrastructure Kit.
- CAVe-Lite Mobile Kit.
- Monitor.
- Keyboard + Mouse.
- Optional: External Computer.

Software Requirements

- V2X Hub

Infrastructure Kit Setup

1. Open a terminal and download the cave-lite GitHub repository if not already downloaded.
 - a. `cd`
 - b. `git clone https://github.com/usdot-fhwa-stol/cave-lite.git`
2. Install the cave-lite dependencies.
 - a. `cd cave-lite/scripts`
 - b. `sudo ./install_dependencies.sh`
3. Configure the V2X Hub SPAT, Immediate Forward, and Message Receiver plugins in the following sections.

V2X Hub + SPAT Plugin

The following plugin configurations are also stated in the System Design Document. These sections provide step-by step instructions on how to make changes to the plugins.

1. Using the Infrastructure Kit computer, login to the V2X Hub UI.
2. Open a web browser and navigate to:
 - a. <http://127.0.0.1>
 - b. **Note:** If using an external computer, you will have to change the IP address and text box to:
 - i. Example: <http://192.168.0.146>
 - ii. Text box example: 192.168.0.146
 - iii. Default text box is 127.0.0.1 if logging in from the Infrastructure Kit computer.
 - iv. Refer to figure 6 for the text box.
3. Login using credentials setup during V2X Hub installation. Example:
 - a. Username: v2xadmin
 - b. Password: V2xHub#321
 - c. Refer to figure 6.



Source: FHWA.

Figure 6. Screenshot. V2X Hub login screen.

4. Once logged in, click on the **SPAT** row to expand it:
 - a. If SPAT is not listed as below, the Enabled, Disabled, and External filters may be toggled on or off to show/hide plugins

Expand the Configuration row and ensure the values match Source: FHWA.

5.
 - a. **Note:** After a change is made to a text box, use <Enter> for the changes to be saved.

W HUB 127.0.0.1
Tue, 09 Apr 2024 18:21:46 GMT

v2xadmin (System Administrator) Sign Out 7.4.0

SPAT 7.5.1

Plugin that reads PTLM data from a configuration file, receives live data from the signal controller, and publishes a J2735 SPA...

Messages

State

Configuration

Key	Value	Default Value	Description
Intersection_Id	1	1	The intersection id for SPAT generated by this plugin.
Intersection_Name	Intersection	Intersection	The intersection name for SPAT generated by this plugin.
Local_IP	127.0.0.1		The IPv4 address of the local computer for receiving Traffic Signal Controller Broadcast Messages.
Local_UDP_Port	6053	local port	The local UDP port for reception of Traffic Signal Controller Broadcast Messages from the TSC.
LogLevel	INFO	INFO	SPAT Plugin Log Level controls which log statements are printed to the terminal.
SignalGroupMapping	<pre>{ "SignalGroups": [{ "SignalGroupId": 1, "Phase": 1, "Type": "vehicle" }, { "SignalGroupId": 2, "Phase": 2, "Type": "vehicle" }, { "SignalGroupId": 3, "Phase": 3, "Type": "vehicle" }, { "SignalGroupId": 4, "Phase": 4, "Type": "vehicle" }, { "SignalGroupId": 5, "Phase": 5, "Type": "vehicle" }, { "SignalGroupId": 6, "Phase": 6, "Type": "vehicle" }, { "SignalGroupId": 7, "Phase": 7, "Type": "vehicle" }, { "SignalGroupId": 8, "Phase": 8, "Type": "vehicle" }, { "SignalGroupId": 22, "Phase": 2, "Type": "pedestrian" }, { "SignalGroupId": 24, "Phase": 4, "Type": "pedestrian" }, { "SignalGroupId": 26, "Phase": 6, "Type": "pedestrian" }, { "SignalGroupId": 28, "Phase": 8, "Type": "pedestrian" }] }</pre>		JSON data defining a list of SignalGroups and phases.
TSC_IP	127.0.0.1		The IPv4 address of the destination Traffic Signal Controller (TSC).
TSC_Remote_SNMP_Port	501		The destination port on the Traffic Signal Controller (TSC) for SNMP NTCIP communication.

Source: FHWA.

Figure 7. Screenshot. SPaT plugin settings.

6. **Enable** the SPAT plugin after the changes have been made.

Immediate Forward

1. Click on the **ImmediateForward** row to expand it.

Expand the Configuration row and ensure the values match the default values. Refer to [Source: FHWA](#).

2. .



127.0.0.1

v2xadmin (System Administrator)
Sign Out
7.4.0

ImmediateForward

7.5.1

Enable

Plugin that listens for TMX messages and forwards them to the V2X Radio (i.e. the RSU).

Remove

Messages

State

Configuration

+

Key	Value	Default Value	Description
Destination_1	127.0.0.1:1516	127.0.0.1:1516	The destination UDP server(s) and port number(s) on the V2X radio for all message s specified by Messages_Destination_1.
Destination_2	0	0	The destination UDP server(s) and port number(s) on the V2X radio for all message s specified by Messages_Destination_2.
Destination_3	0	0	The destination UDP server(s) and port number(s) on the V2X radio for all message s specified by Messages_Destination_3.
Destination_4	0	0	The destination UDP server(s) and port number(s) on the V2X radio for all message s specified by Messages_Destination_4.
HSMurl	http://<softHSM IP>:3000/v1/scms/	http://:3000/v1/scms/	REST API endpoint url needed to make HSM calls
LogLevel	INFO	INFO	The log level for this plugin
Messages_Destination_1	{ "Messages": [{ "TmxType": "SPAT-P", "SendType": "SPA	{ "Messages": [{ "TmxType": "SPAT-P", "SendType": "SPAT", "PSID": "0x8002", "Channel": "183" }, { "TmxType": "MAP-P", "SendType": "MAP", "PSID": "0x8002", "Channel": "183" }, { "TmxType": "PSM-P", "SendType": "PSM", "PSID": "0x27", "Channel": "183" }, { "TmxType": "TMSG07-P", "SendType": "TMSG07", "PSID": "0x8002", "Channel": "183" }, { "TmxType": "TMSG03-P", "SendType": "TMSG03", "PSID": "0xBFE", "Channel": "183" }, { "TmxType": "TMSG05-P", "SendType": "TMSG05", "PSID": "0x8003", "Channel": "183" }, { "TmxType": "SSM-P", "SendType": "SSM", "PSID": "0x8002", "Channel": "183" }, { "TmxType": "SDSM", "SendType": "SensorDataSharingMessage", "PSID": "0x8010", "Channel": "183" }] }	JSON data defining the message types, PSIDs, and channel number for messages forwarded to the V2X radio at destination 1.
	["Messages": []]		JSON data defining the message types, PSIDs, and channel number for messages

Source: FHWA.

Figure 8. Screenshot. ImmediateForward plugin settings.

3. **Enable** the ImmediateFoward plugin.

Message Receiver

1. Click on the **MessageReceiver** row to expand it.

Expand the Configuration row and ensure the values match Source: FHWA.

2. .



Key	Value	Default Value	Description
EnableSimulatedBSM	true	true	Accept and route incoming BSM messages from a V2X Hub simulator.
EnableSimulatedLocation	true	true	Accept and route incoming GPS location messages from a V2X Hub simulator.
EnableSimulatedSRM	true	true	Accept and route incoming SRM messages from a V2X Hub simulator.
EnableVerification	0	0	If enabled, security HSM features are enabled and MessageReceiver Plugin will try to verify incoming message.
HSMUrl	http://<SoftHSM IP>:3000/v1/scms/	http://:3000/v1/scms/	REST API endpoint url needed to make HSM calls
IP	255.255.255.255	127.0.0.1	IP address for the incoming message network connection.
LogLevel	INFO	INFO	The log level for this plugin
messageid	0012,0013,0014,001D,00F0,00F1,00F2,00F3,00F4,00F5,00F6,00F7	0012,0013,0014,001D,00F0,00F1,00F2,00F3,00F4,00F5,00F6,00F7	Comma separated list of SAE J2735 message ID values.
Port	26789	26789	Port for the incoming message network connection.
RouteJ2735	false	false	Set the flag to route/broadcast a received J2735 message to TMX Core.

Source: FHWA.

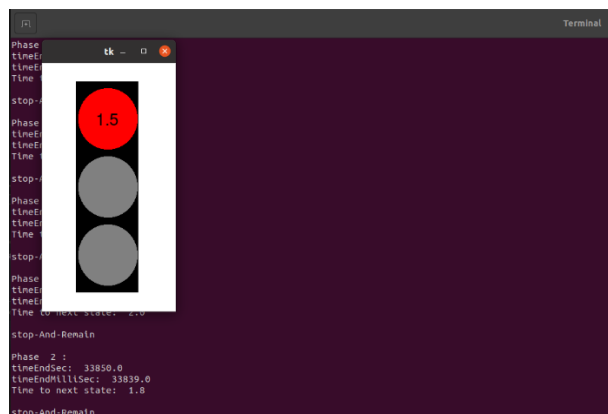
Figure 9. Screenshot. MessageReceiver configuration.

3. Enable the MessageReceiver plugin after changes have been made.

Infrastructure Kit Test

The Infrastructure Kit can be used and tested after the plugin configurations have been set and plugins enabled.

1. `cd ~/cave-lite/src/infrastructure`
2. `sudo ./cave start`
3. You will be prompted to select a file. Type the listed “fullCycle.pcap” and hit <Enter>.
4. A window will open with a traffic signal. Ensure there is a countdown for each signal state. Refer to figure 10.



Source: FHWA.

Figure 10. Screenshot. Traffic signal window.



5. As a final check, you may open the V2X Hub UI and check that the ImmediateForward plugin is sending SPAT messages out to the specified address.
 - a. Expand the ImmediateForward plugin **State tab**.
 - b. Check for an increasing count for SPAT-P. Refer to figure 11.

Filter Plugins: **Enabled** Disabled External Upload File

Plugin	Version	Buttons
CommandPlugin	7.5.1	
ImmediateForward	7.5.1	Disable Remove
Plugin that listens for TMX messages and forwards them to the V2X Radio (i.e. the RSU).		
Messages		
State		
Key	Value	
Messages Skipped (No DSRC metadata)	0	
Messages Skipped (No route)	0	
Messages Skipped (Invalid UDP Client)	0	
Message Skipped (Signature Error Response)	0	
MuteDsdc	false	
Start Time	2024-04-09 18:23:32.403	
SPAT-P	757	
Configuration		
MessageReceiver	7.5.1	Disable
SPAT	7.5.1	Disable

Source: FHWA.

Figure 11. Screenshot. ImmediateForward plugin SPAT-P forwarding.

6. To terminate the scripts, first close the **tk** window then <Ctrl-C> in the active terminal.
7. `sudo ./cave stop`

Mobile Kit Setup

The Mobile Kit listens for broadcasts in the network it is connected to and broadcasts to the same network. It is specifically looking for SAE J2735 SPaT messages. Once SPaT is received, the message is decoded and phase information is extracted. The phase state and timing information is used to control the motor of a 1/10th scale vehicle. The scripts have been defaulted to travel in a straight direction for 50 feet. Once the 50 feet have been travelled, the scripts will terminate.

1. **Using the Mobile Kit computer**, open a terminal and clone the cave-lite repository.
 - a. `cd`
 - b. `git clone https://github.com/usdot-fhwa-stol/cave-lite.git`
2. Install the cave-lite dependencies.
 - a. `cd cave-lite/scripts`
 - b. `sudo ./install_dependencies.sh`

Mobile Kit Test

1. `cd ~/cave-lite/src/mobile`
2. `bash mobile.sh`
3. The script will begin and print out its status until the pre-set destination is reached. Refer to figure 12.



```
File Edit Tabs Help
cave@raspberrypi: ~/cave-lite/src/mobile
State: 2
Current RPM: 240
Current Speed: 0.92
Distance Travelled: 9.15
Vehicle distance to signal: 6.09
Time to next state: 2.6
Phase: protected-Movement-Allowed
State: 2
Current RPM: 240
Current Speed: 0.92
Distance Travelled: 9.21
Vehicle distance to signal: 6.03
Time to next state: 1.8
Phase: protected-Movement-Allowed
State: 2
Current RPM: 240
Current Speed: 0.92
Distance Travelled: 9.46
Vehicle distance to signal: 5.78
Time to next state: 1.5
Phase: protected-Movement-Allowed
State: 2
Current RPM: 240
Current Speed: 0.92
Distance Travelled: 9.9
Vehicle distance to signal: 5.34
Time to next state: 1.5
Phase: protected-Movement-Allowed
State: 2
Current RPM: 240
Current Speed: 0.92
Distance Travelled: 10.72
Vehicle distance to signal: 4.52
Time to next state: 1.5
Phase: protected-Movement-Allowed
State: 2
Current RPM: 240
Current Speed: 0.92
Distance Travelled: 11.92
Vehicle distance to signal: 3.32
Time to next state: 1.2
Phase: protected-Movement-Allowed
State: 2
Current RPM: 240
Current Speed: 0.92
Distance Travelled: 13.69
Vehicle distance to signal: 1.55
Time to next state: 1.6
Phase: protected-Movement-Allowed
State: 2
Current RPM: 240
Current Speed: 0.92
Distance Travelled: 15.69
Vehicle distance to signal: -0.79
Time to next state: 0.8
Vehicle destination reached.
cave@raspberrypi:~/cave-lite/src/mobile $
```

Source: FHWA.

Figure 12. Screenshot. Mobile Kit status output.

- 4. At the Infrastructure Kit, verify the Mobile Kit's BSMs are received.
 - a. Expand the Message Receiver **Messages** tab. Refer to figure 13.

Tue, 09 Apr 2024 19:22:11 GMT7.4.0

PluginsMessagesEvent LogUsers

Filter Plugins

EnabledDisabledExternal

Upload File

CommandPlugin7.5.1

ImmediateForward7.5.1

MessageReceiver7.5.1

Plugin to receive messages from an external V2X radio or other source

Remove

Messages

Type	Subtype	Count	Last Timestamp	Average Interval
J2735	BSM	1721	2024-04-09 19:22:10	57
System	KeepAlive	7	2024-04-09 19:11:28	0
Unknown	Unknown	46	2024-04-03 16:04:22	0

State

Configuration

SPAT7.5.1

Disable

Source: FHWA.

Figure 13. Screenshot. Message Receiver plugin BSM count.

- 5. If desired, the scripts can be terminated early with <Ctrl-C>.



References

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